

Research to Reality: Scaling Generative AI for The Sphere's "Wizard of Oz"

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Figure 1: Research to Reality: Scaling Generative AI for The Sphere's "Wizard of Oz"



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1 Overview

Adapting the 1939 classic *The Wizard of Oz* for the 16K x 16K immersive canvas of the Las Vegas Sphere presented a unique technical paradox: how to translate nearly century-old, 4:3 analog footage into a fully immersive environment. Initiated at a time when viable generative video models did not yet exist, and requiring bespoke solutions beyond the capabilities of emerging commercial tools, this session reveals the collaborative R&D workflow between Magnopus and Google DeepMind.

Panelists will dissect the hybrid production pipeline used to solve the project's most complex challenges: Generative Super Resolution, Character Outpainting, and AI Performance Generation. First, the panel will address the image fidelity challenge. On the Sphere, the original film grain appeared as massive visual noise, and the limited resolution resulted in indistinct, "blobby" shapes. The team will detail the development of novel upscaling architectures designed to fundamentally modernize the footage. We will demonstrate how these custom models suppressed legacy artifacts and synthesized new high-frequency detail, creating sharpness and texture comparable to modern digital cinema, while maintaining the temporal coherence required for large-format playback.

Second, the session explores the integration of generative elements into a traditional VFX environment. While traditional vendor

workflows handled the massive environmental extensions, the team utilized Generative AI to solve the complex character continuity challenges. We will detail the use of Character Outpainting to seamlessly complete characters partially cropped by the original 4:3 frame, alongside the generation of new AI Character Performances for those who walked completely out of view. The team will discuss the strategies used to ensure these generated assets matched the distinct physical language of the original actors.

Finally, the session covers the engineering hurdles of orchestrating this hybrid pipeline. We will examine how the team managed the ingest of disparate high-resolution generative elements to deliver a unified, temporally stable 16K master for the world's largest LED screen.

Since its debut, this pioneering effort has become one of the most commercially successful generative AI deployments in entertainment history, driving record-breaking audiences at Sphere. Validating the success of this hybrid R&D approach, the project earned the prestigious PGA Innovation Award and the VES Award for Outstanding Visual Effects in a Special Venue Project. Attendees will leave with a proven roadmap for navigating the intersection of generative AI research, traditional VFX, and next-generation immersive storytelling.